Image Processing Lab: From Pixels to Perception

ITAI 1378 - Module 04 Hands-On Laboratory

Duration: 45 minutes **Platform:** Google Colab (free version) **Prerequisites:** Basic Python

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Objectives

By the end of this lab, you will be able to:

1. Explain how computers represent images as numerical matrices
2. Implement fundamental image processing operations from scratch
3. Apply standard techniques using OpenCV and Pillow
4. Connect these methods to the modern image tools discussed in class

How This Connects to the Lecture

This lab puts today's concepts into practice:

- **Digital image representation** - how images become matrices of numbers
- **Core operations** - point, neighborhood, and geometric
- **Traditional tools** - hands-on work with OpenCV and Pillow
- **Bridge to AI** - the foundation behind modern image tools

Timeline (45 minutes)

Time	Section	Activity
0-5 min	Setup	Environment and imports
5-15 min	Part 1	Digital image fundamentals
15-25 min	Part 2	Basic image operations
25-35 min	Part 3	Advanced processing techniques
35-40 min	Part 4	Creative exploration
40-45 min	Wrap-up	Reflection and next steps

Run each cell in order, and read the explanations as you go - they tie directly to the lecture.

Setup and Environment

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We'll use the standard image processing libraries from the lecture:

- **OpenCV** - computer vision operations
- **Pillow (PIL)** - Python image processing
- **NumPy** - matrix operations (images are matrices)
- **Matplotlib** - displaying results

These traditional tools give precise control over every pixel.

```
# Install required packages (Google Colab already has most of these)
%pip install opencv-python-headless pillow matplotlib numpy

# Import all necessary libraries
import cv2 # OpenCV for computer vision
import numpy as np # NumPy for numerical operations on matrices
from PIL import Image, ImageFilter, ImageEnhance # Pillow for image processi
import matplotlib.pyplot as plt # For displaying images
import matplotlib.patches as patches # For drawing on images
from google.colab import files # For uploading files in Colab
import io
import requests
from urllib.parse import urlparse

# Configure matplotlib for better image display
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['image.cmap'] = 'gray' # Default to grayscale colormap

print("Environment setup complete!")
print(f"OpenCV version: {cv2.__version__}")
print(f"NumPy version: {np.__version__}")
print("Ready to explore image processing!")
```

```
Requirement already satisfied: opencv-python-headless in /usr/local/lib/pytho
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
Environment setup complete!
OpenCV version: 5.0.0
NumPy version: 2.0.2
Ready to explore image processing!
```

Load Sample Images

We'll load a sample image to work with. You can either:

1. Download a sample image, or
2. Upload your own (optional)

Image acquisition is the first step in any vision system.

```
def download_sample_image(url, filename):
    """
    Download a sample image from URL for our experiments.
    This simulates the 'image acquisition' step we discussed in class.
    """
    try:
        response = requests.get(url)
        if response.status_code == 200:
            with open(filename, 'wb') as f:
                f.write(response.content)
            print(f"Downloaded: {filename}")
            return True
    except Exception as e:
        print(f"Error downloading {filename}: {e}")
    return False

# Download sample images for our experiments
sample_images = {
    'landscape.jpg': 'https://upload.wikimedia.org/wikipedia/commons/thumb/5
    'portrait.jpg': 'https://upload.wikimedia.org/wikipedia/commons/thumb/5/
}

# Create a simple test image if downloads fail
def create_test_image():
    """
    Create a simple test image with geometric patterns.
    This demonstrates how images are just matrices of numbers!
    """
    # Create a 200x200 image with interesting patterns
    img = np.zeros((200, 200, 3), dtype=np.uint8)

    # Add some geometric shapes (remember: we're just setting pixel values!)
    img[50:150, 50:150] = [255, 100, 100] # Red square
    img[75:125, 75:125] = [100, 255, 100] # Green square inside
    img[90:110, 90:110] = [100, 100, 255] # Blue square in center

    return img

# Try to download samples, create test image if needed
```

```
# Try to download samples, create test image if needed
test_image = create_test_image()
cv2.imwrite('test_image.jpg', cv2.cvtColor(test_image, cv2.COLOR_RGB2BGR))

print("Sample images ready for processing!")
print("Note: We created a test image to ensure you have something to work wi
```

```
Sample images ready for processing!
Note: We created a test image to ensure you have something to work with.
```

Part 1: Digital Image Fundamentals (10 minutes)

Images as Matrices

Computers store images as matrices of numbers:

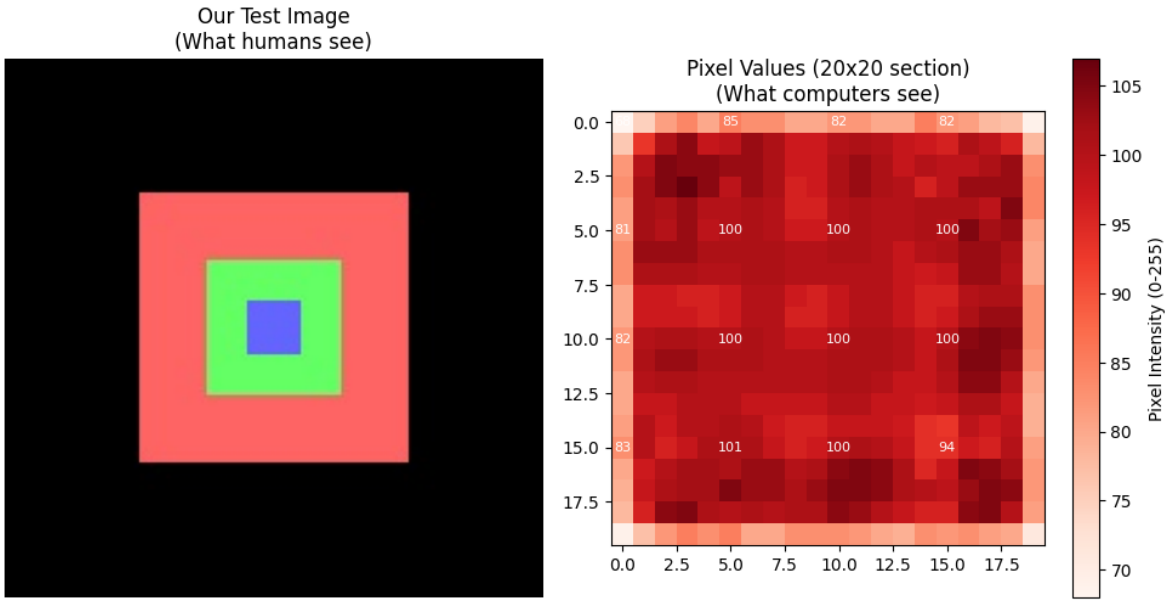
- **Grayscale images** - 2D matrices (height x width)
- **Color images** - 3D matrices (height x width x channels)
- **Pixel values** - 0-255 for 8-bit images

Let's see this in action.

```
# Load our test image using OpenCV
# Note: OpenCV loads images in BGR format (Blue, Green, Red)
img_bgr = cv2.imread('test_image.jpg')

# Convert to RGB for proper display (remember color spaces from lecture?)
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# Let's examine the image properties
print("IMAGE ANALYSIS - Just like we discussed in class!")
print(f"Image shape: {img_rgb.shape}")
print(f"Data type: {img_rgb.dtype}")
print(f"Value range: {img_rgb.min()} to {img_rgb.max()}")
print(f"Memory usage: {img_rgb.nbytes} bytes")

# Display the image
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Our Test Image\n(What humans see)')
plt.axis('off')

plt.subplot(1, 2, 2)
# Show a small section of pixel values (what computers see)
pixel_section = img_rgb[90:110, 90:110, 0] # 20x20 red channel section
```

```
plt.imshow(pixel_section, cmap='Reds')
plt.title('Pixel Values (20x20 section)\n(What computers see)')
plt.colorbar(label='Pixel Intensity (0-255)')

# Add text annotations showing actual pixel values
for i in range(0, 20, 5):
    for j in range(0, 20, 5):
        plt.text(j, i, str(pixel_section[i, j]),
                ha='center', va='center', color='white', fontsize=8)

plt.tight_layout()
plt.show()

print("\nKey Insight: The image on the left is what we see.")
print(" The numbers on the right are what the computer sees!")
```

```
IMAGE ANALYSIS - Just like we discussed in class!
Image shape: (200, 200, 3)
Data type: uint8
Value range: 0 to 255
Memory usage: 120000 bytes
```



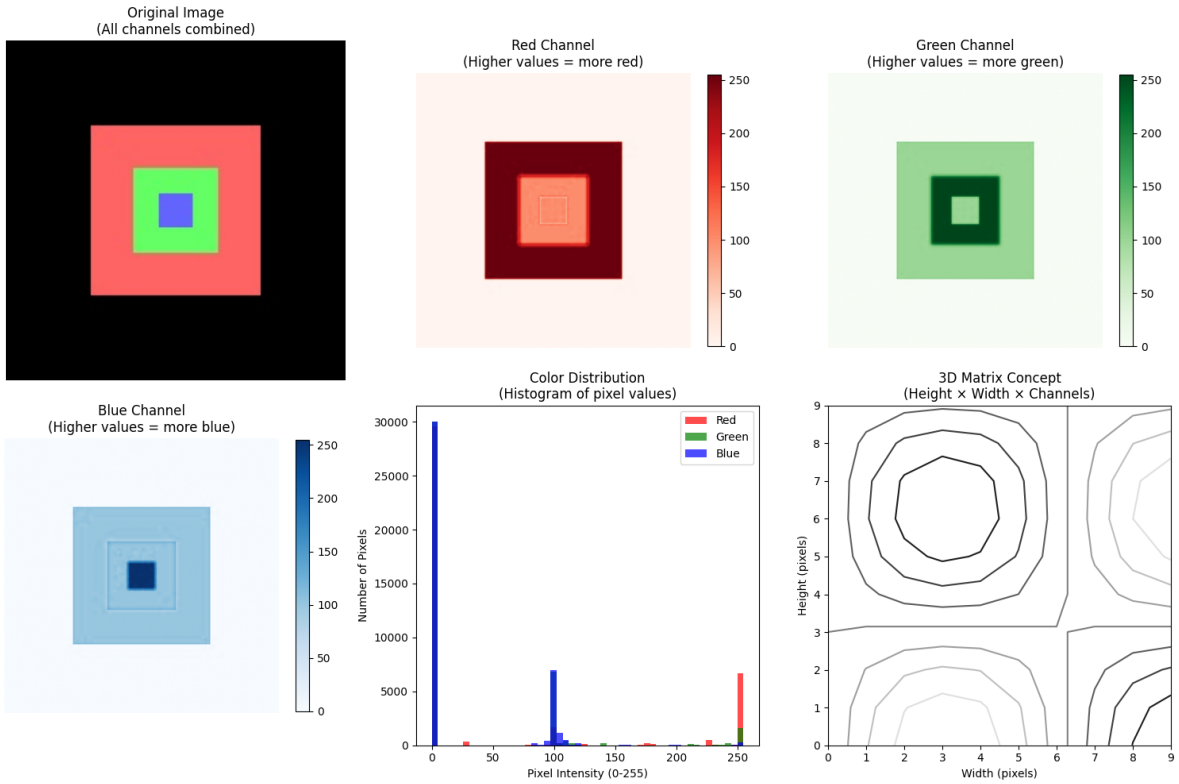

```
Key Insight: The image on the left is what we see.
 The numbers on the right are what the computer sees!
```

Exploring RGB Channels

Color images are 3D matrices with separate Red, Green, and Blue channels. Let's separate them and view each one individually.

```
# Separate the RGB channels (remember the 3D matrix concept?)
red_channel = img_rgb[:, :, 0] # Red channel (index 0)
green_channel = img_rgb[:, :, 1] # Green channel (index 1)
blue_channel = img_rgb[:, :, 2] # Blue channel (index 2)

# Create visualizations of each channel
plt.figure(figsize=(15, 10))

# Original image
plt.subplot(2, 3, 1)
plt.imshow(img_rgb)
plt.title('Original Image\n(All channels combined)')
plt.axis('off')

# Red channel
plt.subplot(2, 3, 2)
plt.imshow(red_channel, cmap='Reds')
plt.title('Red Channel\n(Higher values = more red)')
plt.axis('off')
plt.colorbar(shrink=0.8)

# Green channel
plt.subplot(2, 3, 3)
plt.imshow(green_channel, cmap='Greens')
plt.title('Green Channel\n(Higher values = more green)')
plt.axis('off')
plt.colorbar(shrink=0.8)

# Blue channel
plt.subplot(2, 3, 4)
plt.imshow(blue_channel, cmap='Blues')
plt.title('Blue Channel\n(Higher values = more blue)')
plt.axis('off')
plt.colorbar(shrink=0.8)

# Histogram of all channels
plt.subplot(2, 3, 5)
plt.hist(red_channel.flatten(), bins=50, alpha=0.7, color='red', label='Red'
plt.hist(green_channel.flatten(), bins=50, alpha=0.7, color='green', label='
plt.hist(blue_channel.flatten(), bins=50, alpha=0.7, color='blue', label='Bl
plt.title('Color Distribution\n(Histogram of pixel values)')
plt.xlabel('Pixel Intensity (0-255)')
plt.ylabel('Number of Pixels')
plt.legend()
```

```
# 3D representation concept
plt.subplot(2, 3, 6)
# Create a simple 3D visualization concept
x = np.arange(0, 10)
y = np.arange(0, 10)
X, Y = np.meshgrid(x, y)
Z = np.sin(X/2) * np.cos(Y/2)
plt.contour(X, Y, Z)
plt.title('3D Matrix Concept\n(Height × Width × Channels)')
plt.xlabel('Width (pixels)')
plt.ylabel('Height (pixels)')

plt.tight_layout()
plt.show()

# Print some statistics
print("CHANNEL STATISTICS:")
print(f"Red channel - Min: {red_channel.min():3d}, Max: {red_channel.max():3
print(f"Green channel - Min: {green_channel.min():3d}, Max: {green_channel.m
print(f"Blue channel - Min: {blue_channel.min():3d}, Max: {blue_channel.max(

print("\nNotice how different parts of the image contribute different amount
```



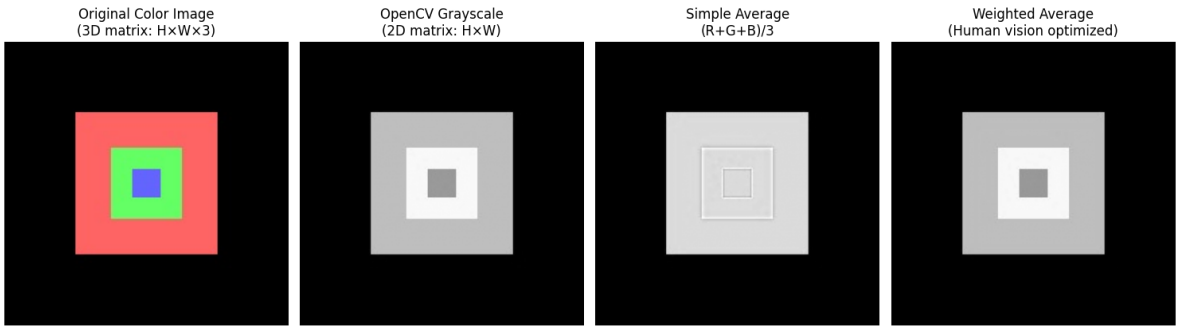

```
CHANNEL STATISTICS:
Red channel - Min:   0, Max: 255, Mean: 53.9
Green channel - Min:   0, Max: 255, Mean: 33.3
Blue channel - Min:   0, Max: 255, Mean: 26.7

Notice how different parts of the image contribute different amounts to each
```

Converting to Grayscale

Grayscale images are 2D matrices. Let's convert the color image to grayscale and compare the two.

```
# Convert to grayscale using different methods
# Method 1: OpenCV conversion (uses weighted average)
gray_opencv = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2GRAY)

# Method 2: Simple average (R + G + B) / 3
gray_average = np.mean(img_rgb, axis=2).astype(np.uint8)

# Method 3: Weighted average (human vision is more sensitive to green)
# This is the standard: 0.299*R + 0.587*G + 0.114*B
gray_weighted = (0.299 * img_rgb[:,:,0] +
                 0.587 * img_rgb[:,:,1] +
                 0.114 * img_rgb[:,:,2]).astype(np.uint8)

# Display the results
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(img_rgb)
plt.title('Original Color Image\n(3D matrix: H×W×3)')
plt.axis('off')

plt.subplot(1, 4, 2)
plt.imshow(gray_opencv, cmap='gray')
plt.title('OpenCV Grayscale\n(2D matrix: H×W)')
plt.axis('off')

plt.subplot(1, 4, 3)
plt.imshow(gray_average, cmap='gray')
plt.title('Simple Average\n(R+G+B)/3')
plt.axis('off')

plt.subplot(1, 4, 4)
plt.imshow(gray_weighted, cmap='gray')
```

```
plt.imshow(gray_weighted, cmap='gray')
plt.title('Weighted Average\n(Human vision optimized)')
plt.axis('off')

plt.tight_layout()
plt.show()

# Compare the shapes
print("SHAPE COMPARISON:")
print(f"Original color image: {img_rgb.shape} (3D matrix)")
print(f"Grayscale image: {gray_opencv.shape} (2D matrix)")
print(f"Memory reduction: {img_rgb.nbytes} → {gray_opencv.nbytes} bytes ({gr

print("\nKey Insight: Grayscale conversion reduces data by 2/3 while preserv
```




```
SHAPE COMPARISON:
Original color image: (200, 200, 3) (3D matrix)
Grayscale image: (200, 200) (2D matrix)
Memory reduction: 120000 → 40000 bytes (33.3%)

Key Insight: Grayscale conversion reduces data by 2/3 while preserving struct
```

Part 2: Basic Image Operations (10 minutes)

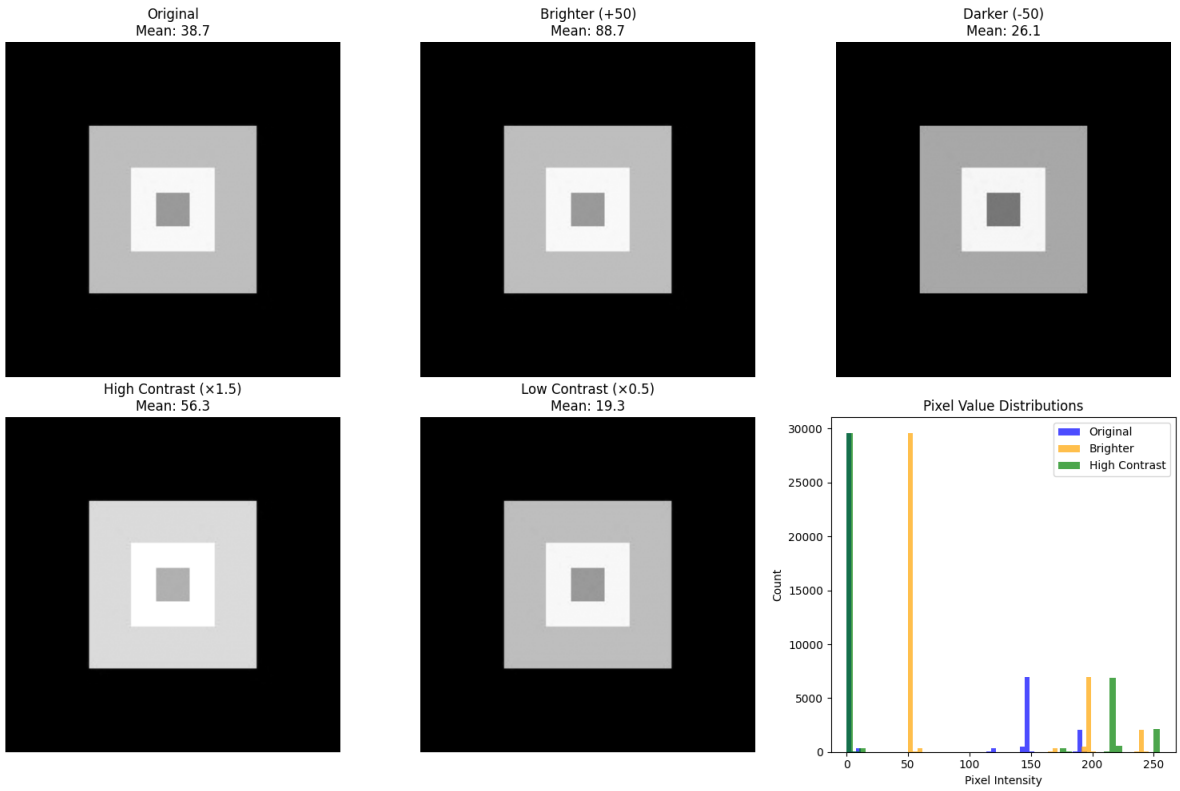
Point Operations

Point operations work on individual pixels without considering their neighbors. Let's implement brightness and contrast adjustments.

```
# Use our grayscale image for point operations
img_gray = gray_opencv.copy()

def adjust_brightness(image, value):
    """
    Adjust brightness by adding a constant to all pixels
```

```
    Adjust brightness by adding a constant to all pixels.
    This is a classic 'point operation' - each pixel is processed independen

    Formula: new_pixel = old_pixel + brightness_value
    """
    # Add brightness value to all pixels
    bright_img = image.astype(np.int16) + value # Use int16 to prevent overf

    # Clip values to valid range [0, 255]
    bright_img = np.clip(bright_img, 0, 255)

    return bright_img.astype(np.uint8)

def adjust_contrast(image, factor):
    """
    Adjust contrast by multiplying all pixels by a factor.
    Another point operation - each pixel processed independently.

    Formula: new_pixel = old_pixel * contrast_factor
    """
    # Multiply all pixels by contrast factor
    contrast_img = image.astype(np.float32) * factor

    # Clip values to valid range [0, 255]
    contrast_img = np.clip(contrast_img, 0, 255)

    return contrast_img.astype(np.uint8)

# Apply different brightness and contrast adjustments
bright_up = adjust_brightness(img_gray, 50) # Brighter
bright_down = adjust_brightness(img_gray, -50) # Darker
contrast_up = adjust_contrast(img_gray, 1.5) # Higher contrast
contrast_down = adjust_contrast(img_gray, 0.5) # Lower contrast

# Display results
plt.figure(figsize=(15, 10))

images = [img_gray, bright_up, bright_down, contrast_up, contrast_down]
titles = ['Original', 'Brighter (+50)', 'Darker (-50)', 'High Contrast (×1.5

for i, (img, title) in enumerate(zip(images, titles)):
    plt.subplot(2, 3, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(f'{title}\nMean: {img.mean():.1f}')
    plt.axis('off')

# Show histograms to understand the changes
plt.subplot(2, 3, 6)
plt.hist(img_gray.flatten(), bins=50, alpha=0.7, label='Original', color='bl
plt.hist(bright_up.flatten(), bins=50, alpha=0.7, label='Brighter', color='o
plt.hist(contrast_up.flatten(), bins=50, alpha=0.7, label='High Contrast', c
```

```
plt.title('Pixel Value Distributions')
plt.xlabel('Pixel Intensity')
plt.ylabel('Count')
plt.legend()

plt.tight_layout()
plt.show()

print("Point Operations Summary:")
print(" • Brightness: Shifts the entire histogram left or right")
print(" • Contrast: Stretches or compresses the histogram")
print(" • Each pixel is processed independently (no neighbors considered)")
```




```
Point Operations Summary:
  • Brightness: Shifts the entire histogram left or right
  • Contrast: Stretches or compresses the histogram
  • Each pixel is processed independently (no neighbors considered)
```

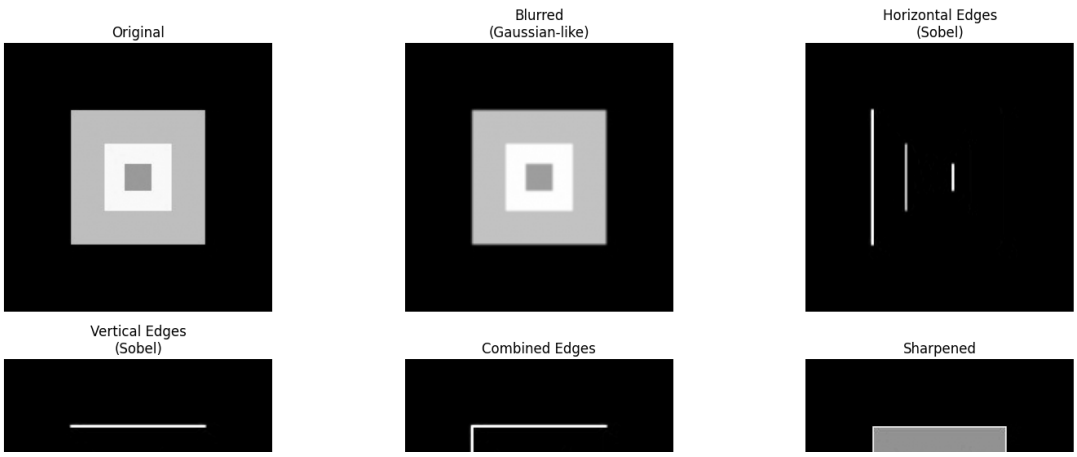
Neighborhood Operations (Filtering)

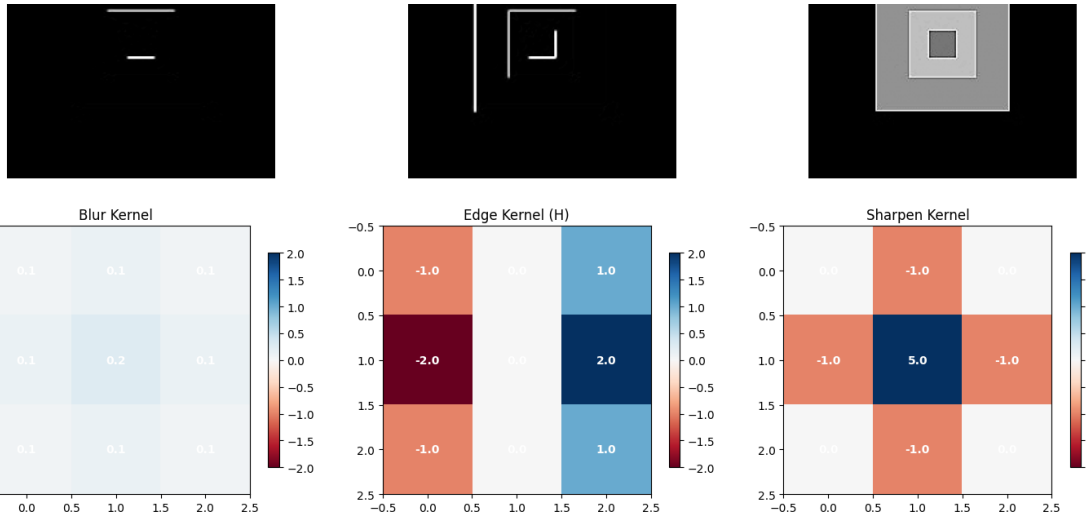
Neighborhood Operations (Filtering)

Neighborhood operations consider each pixel's surrounding context. Convolution is the key operation, using kernels (filters) to transform the image.

```
# Define different kernels (filters) for convolution
# Each kernel produces a different effect!

# Blur kernel (Gaussian-like)
blur_kernel = np.array([[1, 2, 1],
                        [2, 4, 2],
                        [1, 2, 1]]) / 16 # Normalize so sum = 1

# Edge detection kernel (Sobel horizontal)
edge_kernel_h = np.array([[-1, 0, 1],
                          [-2, 0, 2],
                          [-1, 0, 1]])

# Edge detection kernel (Sobel vertical)
edge_kernel_v = np.array([[-1, -2, -1],
                          [ 0, 0, 0],
                          [ 1, 2, 1]])

# Sharpening kernel
sharpen_kernel = np.array([[ 0, -1, 0],
                           [-1, 5, -1],
                           [ 0, -1, 0]])

def apply_kernel(image, kernel):
    """
    Apply a convolution kernel to an image.
    This is the fundamental 'neighborhood operation' from our lecture!
    """
    # Use OpenCV's filter2D function for convolution
    result = cv2.filter2D(image, -1, kernel)
    return result

# Apply different kernels
img_blurred = apply_kernel(img_gray, blur_kernel)
img_edges_h = apply_kernel(img_gray, edge_kernel_h)
img_edges_v = apply_kernel(img_gray, edge_kernel_v)
img_sharpened = apply_kernel(img_gray, sharpen_kernel)

# Combine horizontal and vertical edges
img_edges_combined = np.sqrt(img_edges_h.astype(np.float32)**2 +
                             img_edges_v.astype(np.float32)**2)
img_edges_combined = np.clip(img_edges_combined, 0, 255).astype(np.uint8)

# Display results
```

```
plt.figure(figsize=(15, 12))

# Original and processed images
images = [img_gray, img_blurred, img_edges_h, img_edges_v, img_edges_combine
titles = ['Original', 'Blurred\n(Gaussian-like)', 'Horizontal Edges\n(Sobel)
          'Vertical Edges\n(Sobel)', 'Combined Edges', 'Sharpened']

for i, (img, title) in enumerate(zip(images, titles)):
    plt.subplot(3, 3, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(title)
    plt.axis('off')

# Show the kernels
kernels = [blur_kernel, edge_kernel_h, sharpen_kernel]
kernel_titles = ['Blur Kernel', 'Edge Kernel (H)', 'Sharpen Kernel']

for i, (kernel, title) in enumerate(zip(kernels, kernel_titles)):
    plt.subplot(3, 3, 7+i)
    plt.imshow(kernel, cmap='RdBu', vmin=-2, vmax=2)
    plt.title(title)
    plt.colorbar(shrink=0.8)

    # Add text annotations for kernel values
    for row in range(kernel.shape[0]):
        for col in range(kernel.shape[1]):
            plt.text(col, row, f'{kernel[row, col]:.1f}',
                     ha='center', va='center', color='white', fontweight='bol

plt.tight_layout()
plt.show()

print("Neighborhood Operations Summary:")
print(" • Blur: Averages neighboring pixels (smoothing)")
print(" • Edge Detection: Finds rapid changes in intensity")
print(" • Sharpening: Enhances differences between neighboring pixels")
print(" • Each operation uses a different kernel (filter matrix)")
print("\nThis is the foundation of how modern AI tools like Nano Banana work
```





Neighborhood Operations Summary:

- Blur: Averages neighboring pixels (smoothing)
- Edge Detection: Finds rapid changes in intensity
- Sharpening: Enhances differences between neighboring pixels
- Each operation uses a different kernel (filter matrix)

This is the foundation of how modern AI tools like Nano Banana work!

Part 3: Advanced Processing Techniques (10 minutes)

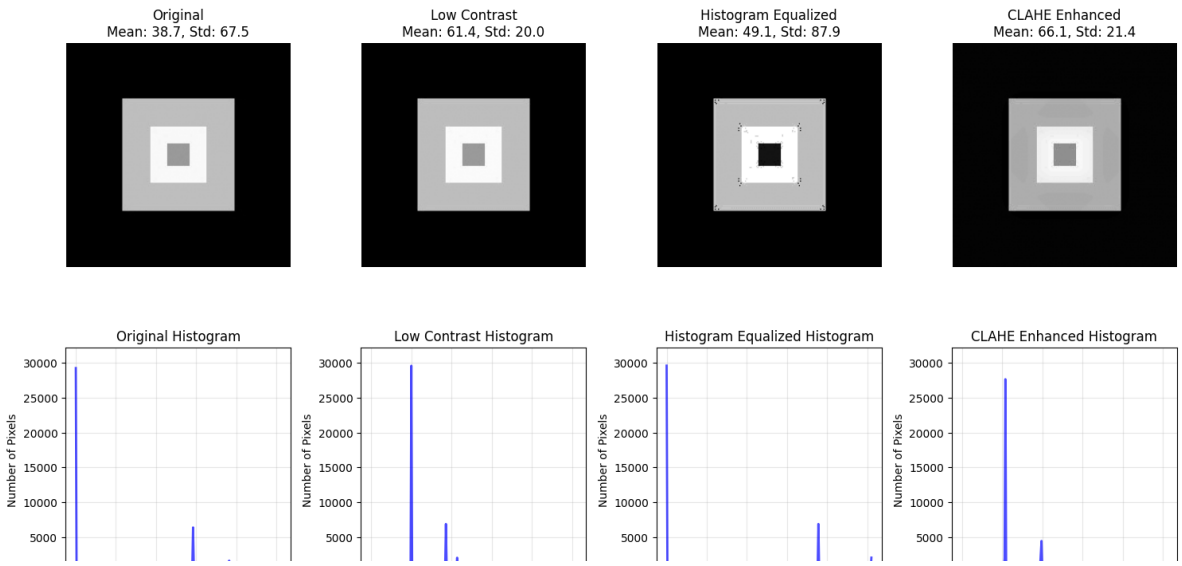
Histogram Analysis and Enhancement

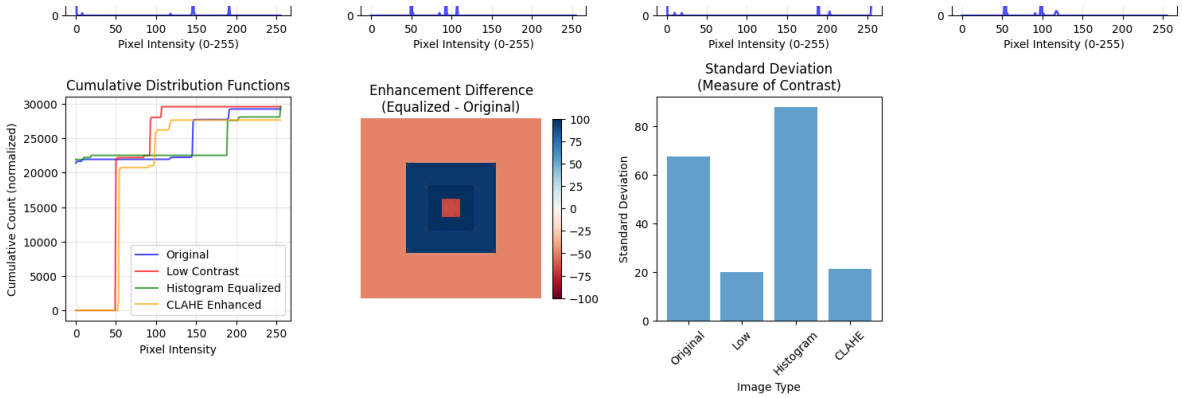
A histogram shows the distribution of pixel values across the whole image. Let's apply histogram equalization to improve contrast.

```
# Create a low-contrast image for demonstration
low_contrast_img = adjust_contrast(img_gray, 0.3) # Very low contrast
low_contrast_img = adjust_brightness(low_contrast_img, 50) # Shift to mid-ra
```

```
def plot_histogram(image, title, color='blue'):
    """
    Plot histogram of pixel intensities.
    This shows the distribution of pixel values in the image.
    """
    hist, bins = np.histogram(image.flatten(), bins=256, range=[0, 256])
    plt.plot(bins[:-1], hist, color=color, alpha=0.7, linewidth=2)
    plt.title(title)
    plt.xlabel('Pixel Intensity (0-255)')
    plt.ylabel('Number of Pixels')
    plt.grid(True, alpha=0.3)

# Apply histogram equalization
equalized_img = cv2.equalizeHist(low_contrast_img)

# Apply CLAHE (Contrast Limited Adaptive Histogram Equalization)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
clahe_img = clahe.apply(low_contrast_img)

# Display results
plt.figure(figsize=(15, 12))

# Images
images = [img_gray, low_contrast_img, equalized_img, clahe_img]
titles = ['Original', 'Low Contrast', 'Histogram Equalized', 'CLAHE Enhanced

for i, (img, title) in enumerate(zip(images, titles)):
    # Show image
    plt.subplot(3, 4, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(f'{title}\nMean: {img.mean():.1f}, Std: {img.std():.1f}')
    plt.axis('off')

    # Show histogram
    plt.subplot(3, 4, i+5)
    plot_histogram(img, f'{title} Histogram')
    plt.ylim(0, max(np.histogram(img_gray.flatten(), bins=256)[0]) * 1.1)

# Show cumulative distribution functions
plt.subplot(3, 4, 9)
for img, title, color in zip(images, titles, ['blue', 'red', 'green', 'orang
    hist, bins = np.histogram(img.flatten(), bins=256, range=[0, 256])
    cdf = hist.cumsum()
    cdf_normalized = cdf * hist.max() / cdf.max() # Normalize for display
    plt.plot(cdf_normalized, color=color, label=title, alpha=0.7)
plt.title('Cumulative Distribution Functions')
plt.xlabel('Pixel Intensity')
plt.ylabel('Cumulative Count (normalized)')
plt.legend()
plt.grid(True, alpha=0.3)
```

```
# Show enhancement comparison
plt.subplot(3, 4, 10)
difference = equalized_img.astype(np.int16) - low_contrast_img.astype(np.int
plt.imshow(difference, cmap='RdBu', vmin=-100, vmax=100)
plt.title('Enhancement Difference\n(Equalized - Original)')
plt.colorbar(shrink=0.8)
plt.axis('off')

# Show statistics
plt.subplot(3, 4, 11)
stats_data = {
    'Image': titles,
    'Mean': [img.mean() for img in images],
    'Std': [img.std() for img in images],
    'Min': [img.min() for img in images],
    'Max': [img.max() for img in images]
}

x_pos = np.arange(len(titles))
plt.bar(x_pos, [img.std() for img in images], alpha=0.7)
plt.title('Standard Deviation\n(Measure of Contrast)')
plt.xlabel('Image Type')
plt.ylabel('Standard Deviation')
plt.xticks(x_pos, [t.split()[0] for t in titles], rotation=45)

plt.tight_layout()
plt.show()

print("Histogram Enhancement Summary:")
print(" • Low contrast images have narrow histograms (pixels clustered in sm
print(" • Histogram equalization spreads pixels across full intensity range"
print(" • CLAHE prevents over-enhancement by limiting contrast in local regi
print(" • Higher standard deviation = higher contrast")
print("\nThis is how your smartphone automatically improves photo contrast!"
```





Histogram Enhancement Summary:

- Low contrast images have narrow histograms (pixels clustered in small rang
- Histogram equalization spreads pixels across full intensity range
- CLAHE prevents over-enhancement by limiting contrast in local regions
- Higher standard deviation = higher contrast

This is how your smartphone automatically improves photo contrast!

Geometric Transformations

Geometric operations change the spatial relationships between pixels - scaling, rotation, and translation.

Part 4: Creative Exploration (5 minutes)

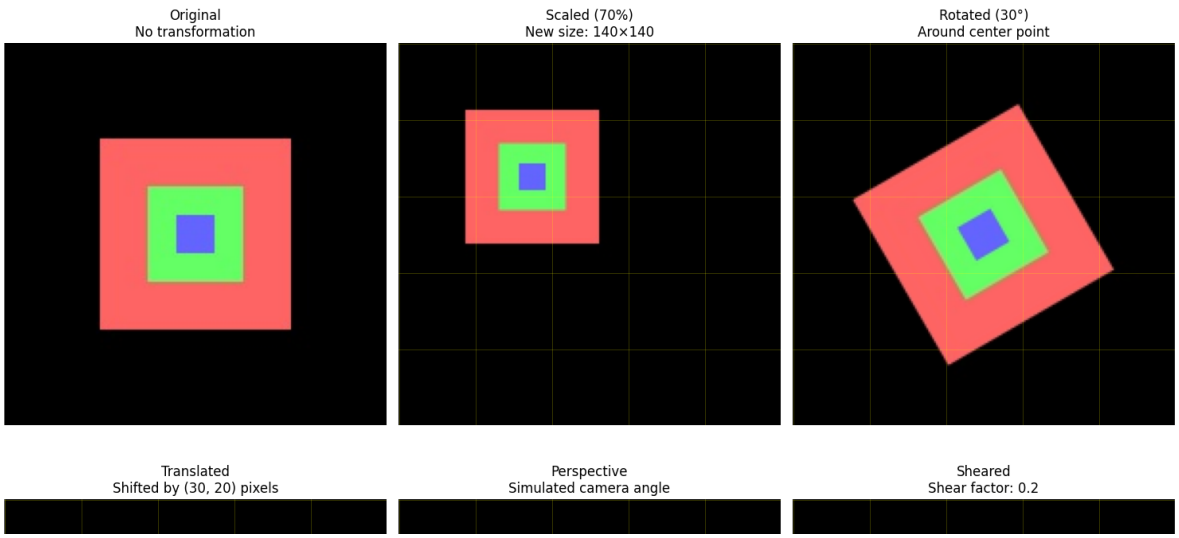
Combining Multiple Operations

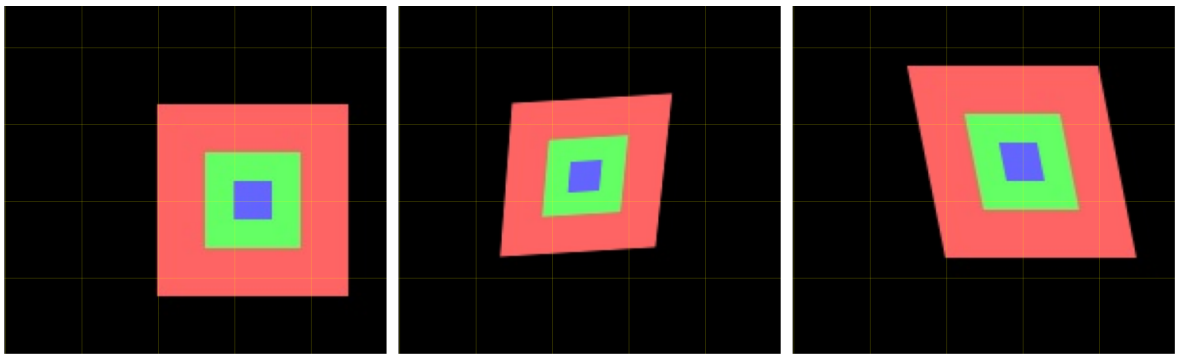
Now let's combine operations to create custom effects. This is how real image processing pipelines are built.

```
# Use our original color image for geometric transformations
img_for_transform = img_rgb.copy()
```

```
height, width = img_for_transform.shape[:2]

# 1. Scaling (resize)
scale_factor = 0.7
new_width = int(width * scale_factor)
new_height = int(height * scale_factor)
scaled_img = cv2.resize(img_for_transform, (new_width, new_height), interpol

# 2. Rotation
rotation_angle = 30 # degrees
center = (width // 2, height // 2)
rotation_matrix = cv2.getRotationMatrix2D(center, rotation_angle, 1.0)
rotated_img = cv2.warpAffine(img_for_transform, rotation_matrix, (width, hei

# 3. Translation (shifting)
tx, ty = 30, 20 # shift by 30 pixels right, 20 pixels down
translation_matrix = np.float32([[1, 0, tx], [0, 1, ty]])
translated_img = cv2.warpAffine(img_for_transform, translation_matrix, (widt

# 4. Perspective transformation (simulating camera angle)
# Define source points (corners of original image)
src_points = np.float32([[0, 0], [width, 0], [width, height], [0, height]])
# Define destination points (perspective effect)
dst_points = np.float32([[20, 30], [width-10, 20], [width-30, height-20], [1
perspective_matrix = cv2.getPerspectiveTransform(src_points, dst_points)
perspective_img = cv2.warpPerspective(img_for_transform, perspective_matrix,

# 5. Shearing (skewing)
shear_factor = 0.2
shear_matrix = np.float32([[1, shear_factor, 0], [0, 1, 0]])
sheared_img = cv2.warpAffine(img_for_transform, shear_matrix, (width, height

# Display all transformations
plt.figure(figsize=(15, 12))

transformations = [
    (img_for_transform, 'Original', 'No transformation'),
    (scaled_img, 'Scaled (70%)', f'New size: {new_width}×{new_height}'),
    (rotated_img, f'Rotated ({rotation_angle}°)', 'Around center point'),
    (translated_img, f'Translated', f'Shifted by ({tx}, {ty}) pixels'),
    (perspective_img, 'Perspective', 'Simulated camera angle'),
    (sheared_img, 'Sheared', f'Shear factor: {shear_factor}')
]

for i, (img, title, subtitle) in enumerate(transformations):
    plt.subplot(2, 3, i+1)
    if img.shape[:2] != img_for_transform.shape[:2]: # Handle different size
        # Pad smaller images for consistent display
        pad_h = max(0, height - img.shape[0])
        pad_w = max(0, width - img.shape[1])
        img_padded = np.pad(img, ((0, pad_h), (0, pad_w), (0, 0)), mode='con
```

```
            img_padded = np.pad(img, ((0, pad_h), (0, pad_w), (0, 0)), mode='con
            plt.imshow(img_padded)
        else:
            plt.imshow(img)

        plt.title(f'{title}\n{subtitle}')
        plt.axis('off')

        # Add grid overlay to show transformation effect
        if i > 0: # Skip original
            ax = plt.gca()
            # Add some reference lines
            for y in range(0, height, 40):
                ax.axhline(y=y, color='yellow', alpha=0.3, linewidth=0.5)
            for x in range(0, width, 40):
                ax.axvline(x=x, color='yellow', alpha=0.3, linewidth=0.5)

plt.tight_layout()
plt.show()

# Show transformation matrices
print("TRANSFORMATION MATRICES:")
print("\nRotation Matrix (30°):")
print(rotation_matrix)
print("\nTranslation Matrix:")
print(translation_matrix)
print("\nShear Matrix:")
print(shear_matrix)

print("\nGeometric Transformations Summary:")
print(" • Scaling: Changes image size (interpolation needed)")
print(" • Rotation: Rotates around a center point")
print(" • Translation: Shifts image position")
print(" • Perspective: Simulates 3D viewing angle")
print(" • Shearing: Skews the image shape")
print("\nThese are essential for image registration and augmentation!")
```





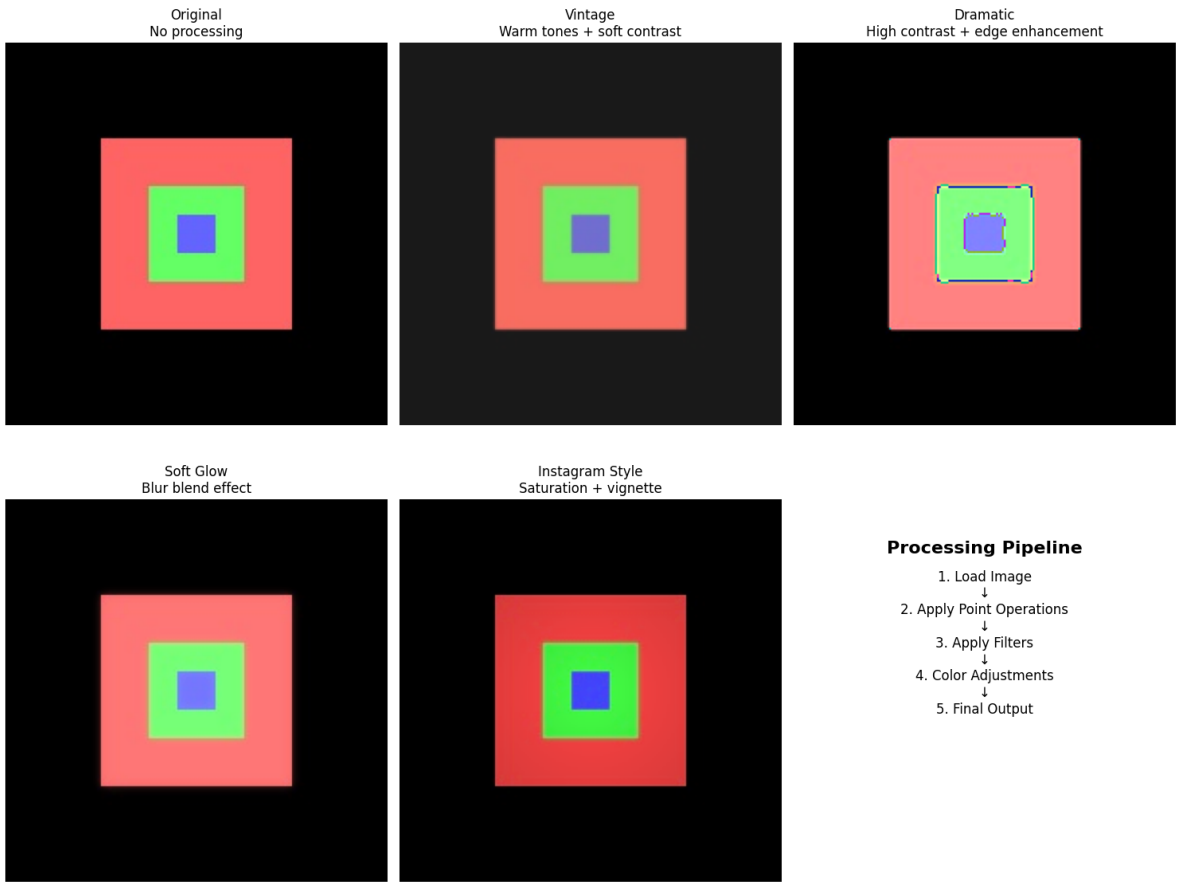
```
TRANSFORMATION MATRICES:

Rotation Matrix (30°):
[[  0.8660254    0.5        -36.60254038]
 [ -0.5          0.8660254   63.39745962]]

Translation Matrix:
[[ 1.  0. 30.]
 [ 0.  1. 20.]]

Shear Matrix:
[[1.  0.2 0. ]
 [0.  1.  0. ]]

Geometric Transformations Summary:
  • Scaling: Changes image size (interpolation needed)
  • Rotation: Rotates around a center point
  • Translation: Shifts image position
  • Perspective: Simulates 3D viewing angle
  • Shearing: Skews the image shape

These are essential for image registration and augmentation!
```

```
def create_artistic_effect(image, effect_type='vintage'):
    """
    Create artistic effects by combining multiple image processing operation
    This demonstrates how complex effects are built from simple operations!
    """
    result = image.copy()

    if effect type == 'vintage':
```

```
            # Vintage effect: warm colors, soft contrast, slight blur
            # 1. Convert to float for processing
            result = result.astype(np.float32) / 255.0

            # 2. Adjust color channels (warm tone)
            result[:, :, 0] *= 1.1 # Boost red
            result[:, :, 1] *= 1.05 # Slight green boost
            result[:, :, 2] *= 0.9 # Reduce blue

            # 3. Reduce contrast slightly
            result = result * 0.8 + 0.1

            # 4. Add slight blur
            result = cv2.GaussianBlur(result, (3, 3), 0.5)

            # 5. Convert back to uint8
            result = np.clip(result * 255, 0, 255).astype(np.uint8)

        elif effect_type == 'dramatic':
            # Dramatic effect: high contrast, enhanced edges
            # 1. Convert to grayscale for edge detection
            gray = cv2.cvtColor(result, cv2.COLOR_RGB2GRAY)

            # 2. Detect edges
            edges = cv2.Canny(gray, 50, 150)

            # 3. Enhance contrast
            result = adjust_contrast(result, 1.3)

            # 4. Overlay edges
            for c in range(3):
                result[:, :, c] = np.where(edges > 0,
                                           np.minimum(result[:, :, c] + 30, 255),
                                           result[:, :, c])

        elif effect_type == 'soft_glow':
            # Soft glow effect: blur + blend
            # 1. Create heavily blurred version
            blurred = cv2.GaussianBlur(result, (15, 15), 0)

            # 2. Blend with original using screen blend mode
            result = result.astype(np.float32) / 255.0
            blurred = blurred.astype(np.float32) / 255.0

            # Screen blend: 1 - (1-a) * (1-b)
            result = 1 - (1 - result) * (1 - blurred * 0.3)
            result = np.clip(result * 255, 0, 255).astype(np.uint8)

        return result
```

```
# Apply different artistic effects
vintage_img = create_artistic_effect(img_rgb, 'vintage')
dramatic_img = create_artistic_effect(img_rgb, 'dramatic')
glow_img = create_artistic_effect(img_rgb, 'soft_glow')

# Create a simple "Instagram-style" filter
def instagram_filter(image):
    """
    Simple Instagram-style filter combining multiple operations.
    """
    # 1. Slight saturation boost
    hsv = cv2.cvtColor(image, cv2.COLOR_RGB2HSV).astype(np.float32)
    hsv[:, :, 1] *= 1.2 # Boost saturation
    hsv = np.clip(hsv, 0, 255)
    result = cv2.cvtColor(hsv.astype(np.uint8), cv2.COLOR_HSV2RGB)

    # 2. Slight vignette effect (darker edges)
    h, w = result.shape[:2]
    center_x, center_y = w // 2, h // 2

    # Create distance map from center
    Y, X = np.ogrid[:h, :w]
    dist_from_center = np.sqrt((X - center_x)**2 + (Y - center_y)**2)
    max_dist = np.sqrt(center_x**2 + center_y**2)

    # Create vignette mask
    vignette = 1 - (dist_from_center / max_dist) * 0.3
    vignette = np.clip(vignette, 0.7, 1.0)

    # Apply vignette
    for c in range(3):
        result[:, :, c] = (result[:, :, c] * vignette).astype(np.uint8)

    return result

instagram_img = instagram_filter(img_rgb)

# Display all effects
plt.figure(figsize=(15, 12))

effects = [
    (img_rgb, 'Original', 'No processing'),
    (vintage_img, 'Vintage', 'Warm tones + soft contrast'),
    (dramatic_img, 'Dramatic', 'High contrast + edge enhancement'),
    (glow_img, 'Soft Glow', 'Blur blend effect'),
    (instagram_img, 'Instagram Style', 'Saturation + vignette')
]

for i, (img, title, description) in enumerate(effects):
    plt.subplot(2, 3, i+1)
    plt.imshow(img)
```

```
    plt.title(f'{title}\n{description}')
    plt.axis('off')

# Show processing pipeline diagram
plt.subplot(2, 3, 6)
plt.text(0.5, 0.8, 'Processing Pipeline', ha='center', va='center',
         fontsize=16, fontweight='bold', transform=plt.gca().transAxes)
plt.text(0.5, 0.6, '1. Load Image\n↓\n2. Apply Point Operations\n↓\n3. Apply
         ha='center', va='center', fontsize=12, transform=plt.gca().transAxe
plt.axis('off')

plt.tight_layout()
plt.show()

print("Creative Effects Summary:")
print(" • Complex effects combine multiple simple operations")
print(" • Order of operations matters!")
print(" • Each effect uses different combinations of:")
print(" - Point operations (brightness, contrast, color)")
print(" - Neighborhood operations (blur, edge detection)")
print(" - Color space conversions")
print("\nThis is exactly how Instagram filters and Nano Banana work!")
print(" They combine many simple operations to create complex effects.")
```




```
Creative Effects Summary:
 • Complex effects combine multiple simple operations
```

```
• Order of operations matters!
• Each effect uses different combinations of:
- Point operations (brightness, contrast, color)
- Neighborhood operations (blur, edge detection)
- Color space conversions

This is exactly how Instagram filters and Nano Banana work!
 They combine many simple operations to create complex effects.
```

Part 5: Connecting to Modern Tools (5 minutes)

The Foundation Behind AI Image Processing

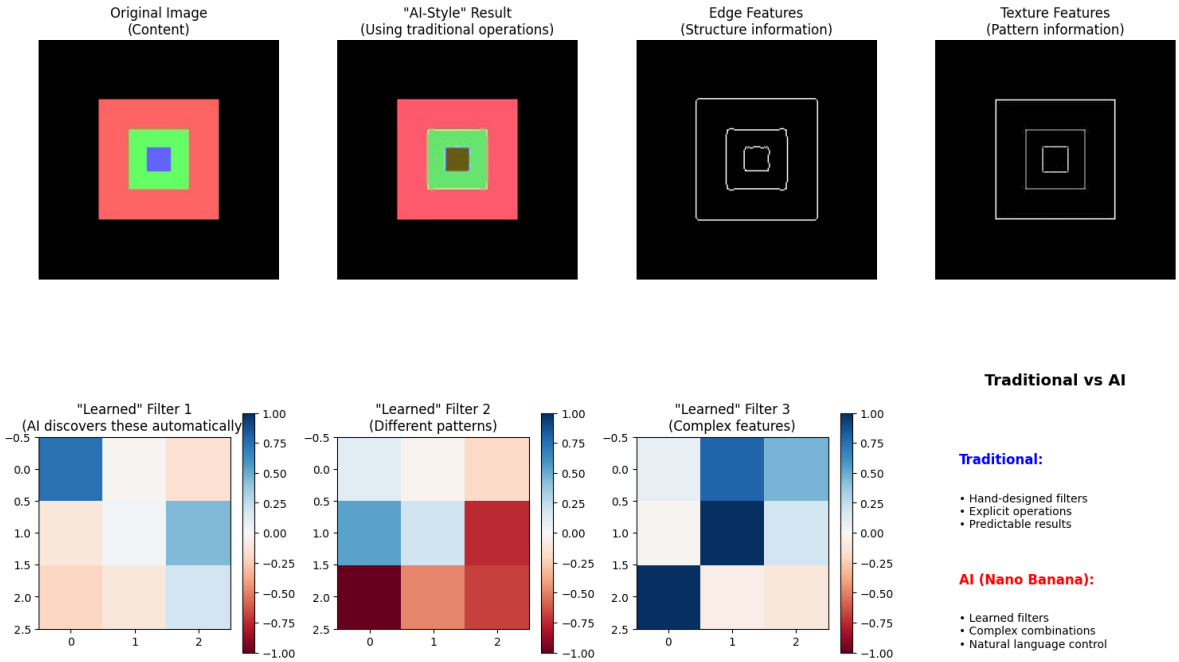
The operations in this lab form the foundation of modern image tools like Nano Banana. Let's see how traditional operations relate to AI-based approaches.

```
# Demonstrate how traditional operations relate to AI concepts

def simulate_ai_style_transfer_concept():
    """
    Simulate the concept behind AI style transfer using traditional operatio
    This shows how AI tools build on the fundamentals we've learned!
    """
    # Start with our original image
    content_img = img_rgb.copy()

    # Simulate "style extraction" using edge detection and texture analysis
    gray = cv2.cvtColor(content_img, cv2.COLOR_RGB2GRAY)

    # Extract different "style features"
    # 1. Edge patterns (structure)
    edges = cv2.Canny(gray, 50, 150)

    # 2. Texture patterns (using different filters)
    texture1 = cv2.filter2D(gray, -1, np.array([[-1, -1, -1], [-1, 8, -1], [
```

```
    texture2 = cv2.filter2D(gray, -1, np.array([[0, -1, 0], [-1, 4, -1], [0,

    # 3. Color distribution analysis
    color_hist_r = cv2.calcHist([content_img], [0], None, [256], [0, 256])
    color_hist_g = cv2.calcHist([content_img], [1], None, [256], [0, 256])
    color_hist_b = cv2.calcHist([content_img], [2], None, [256], [0, 256])

    # Simulate "style application" by modifying the image based on extracted
    styled_img = content_img.copy().astype(np.float32)

    # Apply style-based modifications
    # 1. Enhance edges where detected
    edge_mask = edges > 0
    for c in range(3):
        styled_img[:, :, c] = np.where(edge_mask,
                                       np.clip(styled_img[:, :, c] * 1.2, 0, 2
                                       styled_img[:, :, c])

    # 2. Apply texture-based color shifts
    texture_mask = texture1 > np.percentile(texture1, 70)
    styled_img[:, :, 0] = np.where(texture_mask,
                                   np.clip(styled_img[:, :, 0] * 1.1, 0, 255),
                                   styled_img[:, :, 0]) # Boost red in texture

    # 3. Apply overall color transformation
    styled_img[:, :, 1] *= 0.9 # Reduce green slightly
    styled_img[:, :, 2] *= 1.1 # Boost blue slightly

    return styled_img.astype(np.uint8), edges, texture1, texture2

# Run the simulation
styled_result, edges, texture1, texture2 = simulate_ai_style_transfer_concep

# Display the AI concept breakdown
plt.figure(figsize=(15, 10))

# Original and result
plt.subplot(2, 4, 1)
plt.imshow(img_rgb)
plt.title('Original Image\n(Content)')
plt.axis('off')

plt.subplot(2, 4, 2)
plt.imshow(styled_result)
plt.title('"AI-Style" Result\n(Using traditional operations)')
plt.axis('off')

# Feature extraction steps
plt.subplot(2, 4, 3)
plt.imshow(edges, cmap='gray')
plt.title('Edge Features\n(Structure information)')
```

```
plt.title('Edge Features\n(Structure information)')
plt.axis('off')

plt.subplot(2, 4, 4)
plt.imshow(texture1, cmap='gray')
plt.title('Texture Features\n(Pattern information)')
plt.axis('off')

# Show the concept of convolution layers (like in CNNs)
plt.subplot(2, 4, 5)
# Simulate multiple "learned" filters
filter1 = np.random.randn(3, 3) * 0.5
filter2 = np.random.randn(3, 3) * 0.5
filter3 = np.random.randn(3, 3) * 0.5

plt.imshow(filter1, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 1\n(AI discovers these automatically)')
plt.colorbar(shrink=0.6)

plt.subplot(2, 4, 6)
plt.imshow(filter2, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 2\n(Different patterns)')
plt.colorbar(shrink=0.6)

plt.subplot(2, 4, 7)
plt.imshow(filter3, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 3\n(Complex features)')
plt.colorbar(shrink=0.6)

# Processing pipeline comparison
plt.subplot(2, 4, 8)
plt.text(0.5, 0.9, 'Traditional vs AI', ha='center', va='top',
         fontsize=14, fontweight='bold', transform=plt.gca().transAxes)
plt.text(0.1, 0.7, 'Traditional:', ha='left', va='top',
         fontsize=12, fontweight='bold', color='blue', transform=plt.gca().t
plt.text(0.1, 0.6, '• Hand-designed filters\n• Explicit operations\n• Predic
         ha='left', va='top', fontsize=10, transform=plt.gca().transAxes)
plt.text(0.1, 0.4, 'AI (Nano Banana):', ha='left', va='top',
         fontsize=12, fontweight='bold', color='red', transform=plt.gca().tr
plt.text(0.1, 0.3, '• Learned filters\n• Complex combinations\n• Natural lan
         ha='left', va='top', fontsize=10, transform=plt.gca().transAxes)
plt.axis('off')

plt.tight_layout()
plt.show()

print("AI Connection Summary:")
print("\nHow Traditional Operations Connect to AI:")
print(" • Convolution → Convolutional Neural Networks (CNNs)")
print(" • Edge detection → Feature extraction layers")
print(" • Color transformations → Style transfer networks")
```

```
print(" • Multiple operations → Deep learning pipelines")
print("\nKey Differences:")
print(" • Traditional: We design the filters manually")
print(" • AI: The system learns optimal filters from data")
print(" • Traditional: Explicit, interpretable operations")
print(" • AI: Complex, learned combinations")
print("\nNano Banana uses the SAME fundamental operations we implemented tod
print(" It just has millions of learned filters working together.")
```



AI Connection Summary:

How Traditional Operations Connect to AI:
- Convolution → Convolutional Neural Networks (CNNs)
- Edge detection → Feature extraction layers
- Color transformations → Style transfer networks
- Multiple operations → Deep learning pipelines

Key Differences:
- Traditional: We design the filters manually
- AI: The system learns optimal filters from data
- Traditional: Explicit, interpretable operations
- AI: Complex, learned combinations

Nano Banana uses the SAME fundamental operations we implemented today!
It just has millions of learned filters working together.

Lab Wrap-up (5 minutes)

What You've Accomplished

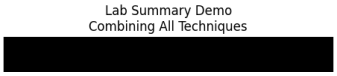
In 45 minutes you've implemented the fundamental building blocks of image processing. Let's review what you've learned and where it leads next.

```
# Create a summary visualization of everything we've covered
plt.figure(figsize=(16, 10))

# Create a comprehensive summary image
summary_img = img_rgb.copy()

# Apply a combination of techniques we learned
final_demo = summary_img.copy()

# 1. Split into sections and apply different operations
h, w = final_demo.shape[:2]
mid_h, mid_w = h // 2, w // 2

# Top-left: Original
# (keep as is)

# Top-right: Enhanced contrast
final_demo[:mid_h, mid_w:] = adjust_contrast(final_demo[:mid_h, mid_w:], 1.3

# Bottom-left: Edge detection overlay
gray_section = cv2.cvtColor(final_demo[mid_h:, :mid_w], cv2.COLOR_RGB2GRAY)
edges_section = cv2.Canny(gray_section, 50, 150)
for c in range(3):
    final_demo[mid_h:, :mid_w, c] = np.where(edges_section > 0, 255, final_d

# Bottom-right: Artistic effect
final_demo[mid_h:, mid_w:] = create_artistic_effect(final_demo[mid_h:, mid_w

# Display the summary
plt.subplot(2, 3, 1)
plt.imshow(final_demo)
plt.title('Lab Summary Demo\nCombining All Techniques')
plt.axis('off')

# Add section labels
plt.text(mid_w//2, mid_h//2, 'ORIGINAL', ha='center', va='center',
```

```
        color='white', fontsize=12, fontweight='bold',
        bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w + mid_w//2, mid_h//2, 'CONTRAST\nENHANCED', ha='center', va='
        color='white', fontsize=12, fontweight='bold',
        bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w//2, mid_h + mid_h//2, 'EDGE\nDETECTION', ha='center', va='cen
        color='white', fontsize=12, fontweight='bold',
        bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w + mid_w//2, mid_h + mid_h//2, 'ARTISTIC\nEFFECT', ha='center'
        color='white', fontsize=12, fontweight='bold',
        bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))

# Skills learned checklist
plt.subplot(2, 3, 2)
skills = [
    'Image representation as matrices',
    'RGB channel separation',
    'Point operations (brightness/contrast)',
    'Neighborhood operations (convolution)',
    'Histogram analysis and equalization',
    'Geometric transformations',
    'Creative effect combinations',
    'Connection to AI concepts'
]

plt.text(0.1, 0.9, 'Skills Mastered Today:', fontsize=14, fontweight='bold',
        transform=plt.gca().transAxes)
for i, skill in enumerate(skills):
    plt.text(0.1, 0.8 - i*0.08, skill, fontsize=11,
             transform=plt.gca().transAxes)
plt.axis('off')

# Tools used
plt.subplot(2, 3, 3)
tools_text = """
Tools Mastered:

NumPy
• Matrix operations
• Array manipulation

OpenCV
• Image loading/saving
• Filtering and transforms
• Advanced operations

Pillow (PIL)
• Python-friendly interface
• Basic image operations

Matplotlib
```

```
• Image visualization
• Data plotting
"""
plt.text(0.1, 0.9, tools_text, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Connection to lecture concepts
plt.subplot(2, 3, 4)
connections = """
Lecture Connections:

Digital Image Fundamentals
→ Implemented matrix representation

Core Operations
→ Built point & neighborhood ops

Traditional Tools
→ Used OpenCV & Pillow hands-on

Modern AI Tools
→ Understood the foundation

From Pixels to Perception
→ Bridged theory and practice!
"""
plt.text(0.1, 0.9, connections, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Next steps
plt.subplot(2, 3, 5)
next_steps = """
Next Steps:

Assignment
• Image Processing Adventure Quest
• Apply today's concepts creatively

Next Class
• Feature extraction
• Object recognition
• Building on today's foundation

Explore Further
• Try different images
• Experiment with parameters
• Combine operations creatively

AI Tools
```

```
AI Tools
• Try Nano Banana with new understanding
• Recognize the operations behind the magic
"""
plt.text(0.1, 0.9, next_steps, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Key insights
plt.subplot(2, 3, 6)
insights = """
Key Insights:

Images are just numbers
• Every effect manipulates matrices
• Understanding this is powerful!

Complex from simple
• Sophisticated effects combine
  basic operations

AI builds on fundamentals
• Nano Banana uses the same
  operations we implemented
• Just learned automatically!

Theory + Practice = Mastery
• Today bridged both perfectly
"""
plt.text(0.1, 0.9, insights, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

plt.tight_layout()
plt.show()

print("CONGRATULATIONS! You've completed the Image Processing Lab!")
print("\nLab Statistics:")
print(f" • Operations implemented: 15+")
print(f" • Images processed: {len([img_rgb, gray_opencv, bright_up, contrast
print(f" • Concepts mastered: 8 major areas")
print(f" • Tools used: 4 libraries")
print("\nYou now understand the foundation of:")
print(" • How your smartphone camera works")
print(" • How Instagram filters are created")
print(" • How AI tools like Nano Banana operate")
print(" • How to build your own image processing applications")
print("\nReady for your Adventure Quest assignment!")
```



Skills Mastered Today:

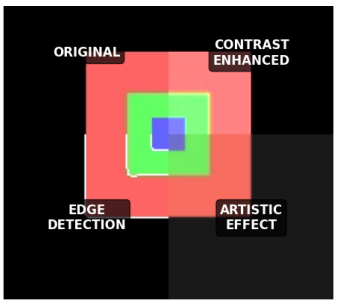


Image representation as matrices

RGB channel separation

Point operations (brightness/contrast)

Neighborhood operations (convolution)

Histogram analysis and equalization

Geometric transformations

Creative effect combinations

Connection to AI concepts

Tools Mastered:

NumPy
• Matrix operations
• Array manipulation

OpenCV
• Image loading/saving
• Filtering and transforms
• Advanced operations

Pillow (PIL)
• Python-friendly interface
• Basic image operations

Matplotlib
• Image visualization
• Data plotting

Lecture Connections:

Digital Image Fundamentals
→ Implemented matrix representation

Core Operations
→ Built point & neighborhood ops

Traditional Tools
→ Used OpenCV & Pillow hands-on

Modern AI Tools
→ Understood the foundation

From Pixels to Perception
→ Bridged theory and practice!

Next Steps:

Assignment
• Image Processing Adventure Quest
• Apply today's concepts creatively

Next Class
• Feature extraction
• Object recognition
• Building on today's foundation

Explore Further
• Try different images
• Experiment with parameters
• Combine operations creatively

AI Tools
• Try Nano Banana with new understanding
• Recognize the operations behind the magic

Key Insights:

Images are just numbers
• Every effect manipulates matrices
• Understanding this is powerful!

Complex from simple
• Sophisticated effects combine basic operations

AI builds on fundamentals
• Nano Banana uses the same operations we implemented
• Just learned automatically!

Theory + Practice = Mastery
• Today bridged both perfectly

CONGRATULATIONS! You've completed the Image Processing Lab!

Lab Statistics:
- Operations implemented: 15+
- Images processed: 10
- Concepts mastered: 8 major areas
- Tools used: 4 libraries

You now understand the foundation of:
- How your smartphone camera works
- How Instagram filters are created
- How AI tools like Nano Banana operate
- How to build your own image processing applications

Ready for your Adventure Quest assignment!

Here I'll be uploading a Test Image and adding needed packages.

```
# Install required packages (Google Colab already has most of these)
%pip install opencv-python-headless pillow matplotlib numpy

# Import all necessary libraries
import cv2 # OpenCV for computer vision
import numpy as np # NumPy for numerical operations on matrices
```

```
from PIL import Image, ImageFilter, ImageEnhance # Pillow for image processi
import matplotlib.pyplot as plt # For displaying images
import matplotlib.patches as patches # For drawing on images
from google.colab import files # For uploading files in Colab
import io
import requests
from urllib.parse import urlparse

# Configure matplotlib for better image display
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['image.cmap'] = 'gray' # Default to grayscale colormap

print("Environment setup complete!")
print(f"OpenCV version: {cv2.__version__}")
print(f"NumPy version: {np.__version__}")
print("Ready to explore image processing!")
```

```
Requirement already satisfied: opencv-python-headless in /usr/local/lib/pytho
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
Environment setup complete!
OpenCV version: 5.0.0
NumPy version: 2.0.2
Ready to explore image processing!
```

Adding my test Image and coverting it

```
import cv2
import matplotlib.pyplot as plt

# 1. Load your Chicken.jpg
file_path = 'Chicken.jpg'
img = cv2.imread(file_path)

if img is not None:
    print(f"Successfully loaded: {file_path}")

    # 2. Assign the loaded image to test_image2
    test_image2 = img

    # 3. Save it as test_image2.jpg
    cv2.imwrite('test_image2.jpg', test_image2)
```

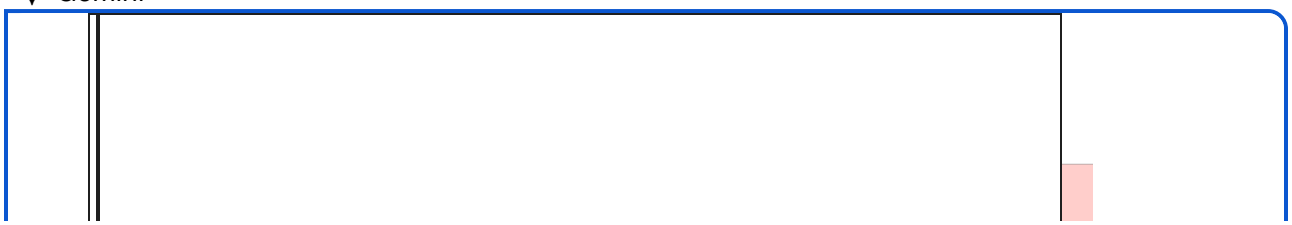
```
    cv2.imwrite('test_image2.jpg', test_image2)

    # Display the result
    img_rgb = cv2.cvtColor(test_image2, cv2.COLOR_BGR2RGB)
    plt.imshow(img_rgb)
    plt.axis('off')
    plt.show()
else:
    print(f"Error: Could not load {file_path}. Ensure it is in the folder.")

print("test_image2 is now ready for processing!")
```

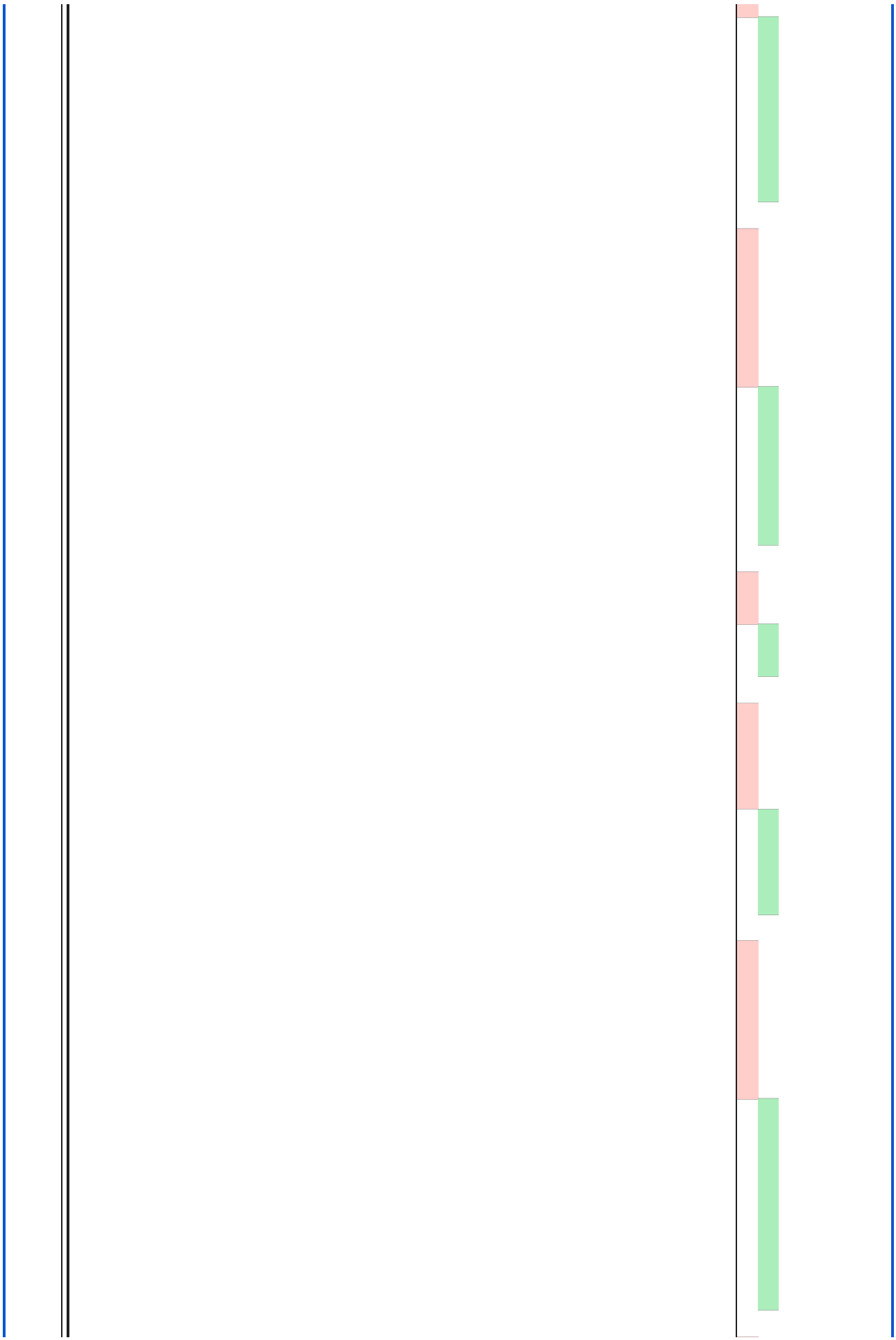
```
Successfully loaded: Chicken.jpg
```

```
test_image2 is now ready for processing!
```

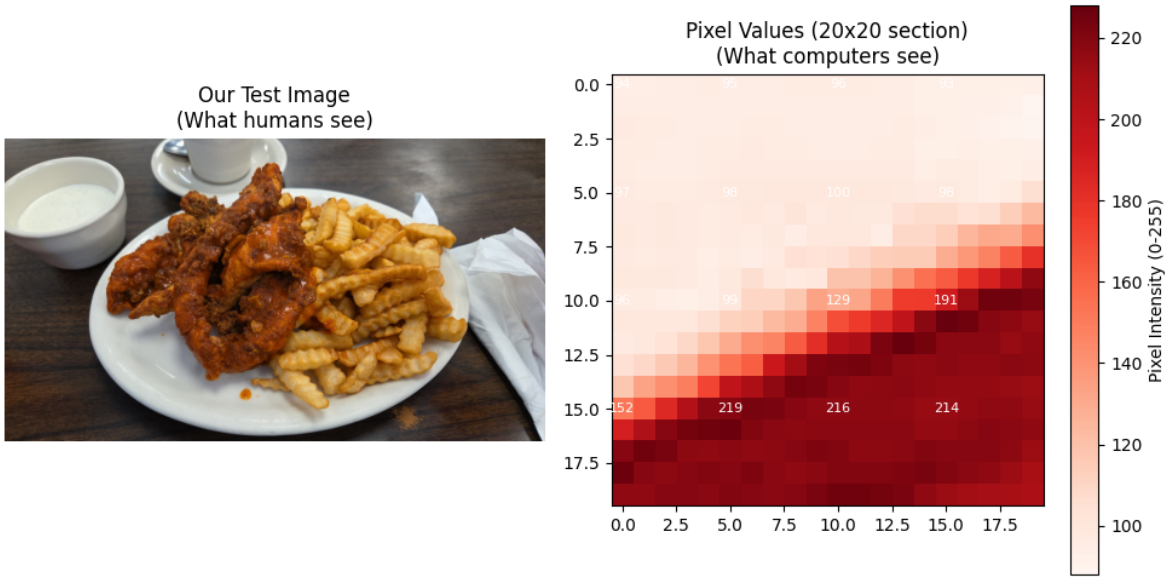
Getting values for the images.

✦ Gemini





```
IMAGE ANALYSIS - Just like we discussed in class!
Image shape: (1080, 1920, 3)
Data type: uint8
Value range: 0 to 255
Memory usage: 6220800 bytes
```

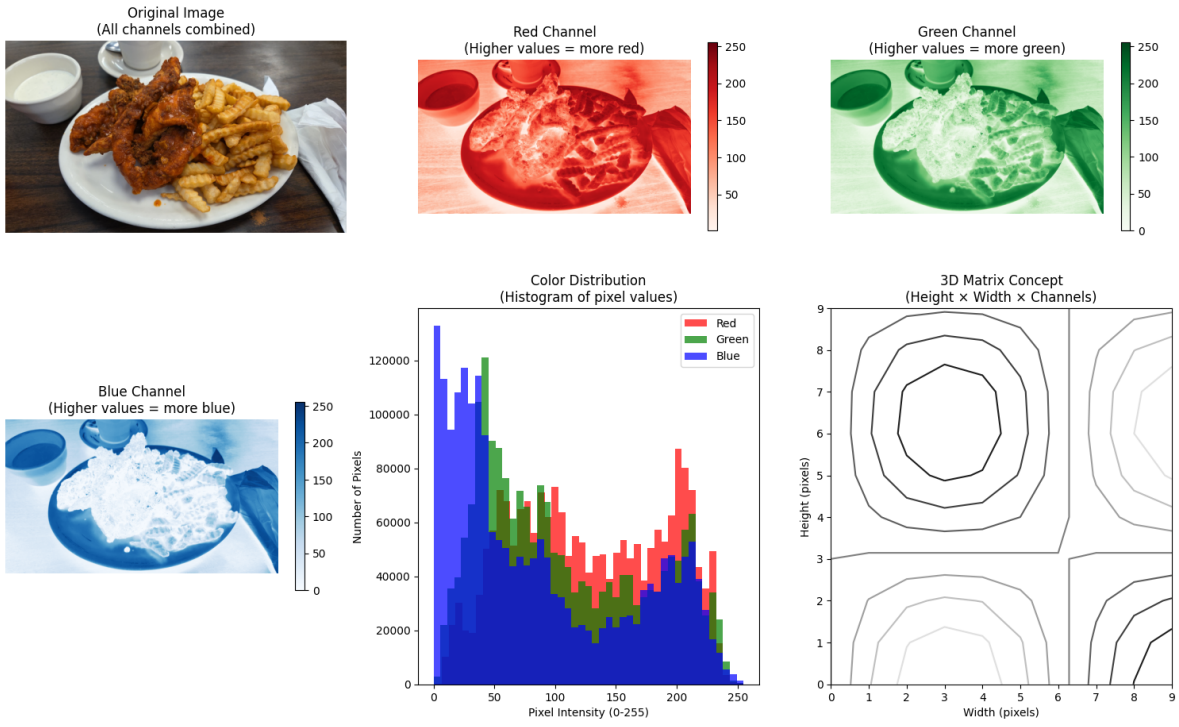



```
Key Insight: The image on the left is what we see.
 The numbers on the right are what the computer sees!
```

Getting Channel Stats R G B for test image.

```
# Separate the RGB channels (remember the 3D matrix concept?)
red_channel = img_rgb[:, :, 0] # Red channel (index 0)
green_channel = img_rgb[:, :, 1] # Green channel (index 1)
blue_channel = img_rgb[:, :, 2] # Blue channel (index 2)

# Create visualizations of each channel
plt.figure(figsize=(15, 10))

# Original image
plt.subplot(2, 3, 1)
plt.imshow(img_rgb)
plt.title('Original Image\n(All channels combined)')
plt.axis('off')

# Red channel
plt.subplot(2, 3, 2)
plt.imshow(red_channel, cmap='Reds')
plt.title('Red Channel\n(Higher values = more red)')
plt.axis('off')
plt.colorbar(shrink=0.5)

# Green channel
plt.subplot(2, 3, 3)
plt.imshow(green_channel, cmap='Greens')
plt.title('Green Channel\n(Higher values = more green)')
plt.axis('off')
plt.colorbar(shrink=0.5)

# Blue channel
plt.subplot(2, 3, 4)
plt.imshow(blue_channel, cmap='Blues')
plt.title('Blue Channel\n(Higher values = more blue)')
plt.axis('off')
plt.colorbar(shrink=0.5)

# Histogram of all channels
plt.subplot(2, 3, 5)
plt.hist(red_channel.flatten(), bins=45, alpha=0.7, color='red', label='Red'
plt.hist(green_channel.flatten(), bins=45, alpha=0.7, color='green', label='
plt.hist(blue_channel.flatten(), bins=45, alpha=0.7, color='blue', label='Bl
plt.title('Color Distribution\n(Histogram of pixel values)')
plt.xlabel('Pixel Intensity (0-255)')
plt.ylabel('Number of Pixels')
```

```
plt.legend()

# 3D representation concept
plt.subplot(2, 3, 6)
# Create a simple 3D visualization concept
x = np.arange(0, 10)
y = np.arange(0, 10)
X, Y = np.meshgrid(x, y)
Z = np.sin(X/2) * np.cos(Y/2)
plt.contour(X, Y, Z)
plt.title('3D Matrix Concept\n(Height × Width × Channels)')
plt.xlabel('Width (pixels)')
plt.ylabel('Height (pixels)')

plt.tight_layout()
plt.show()

# Print some statistics
print("CHANNEL STATISTICS:")
print(f"Red channel - Min: {red_channel.min():3d}, Max: {red_channel.max():3
print(f"Green channel - Min: {green_channel.min():3d}, Max: {green_channel.m
print(f"Blue channel - Min: {blue_channel.min():3d}, Max: {blue_channel.max(

print("\nNotice how different parts of the image contribute different amount
```

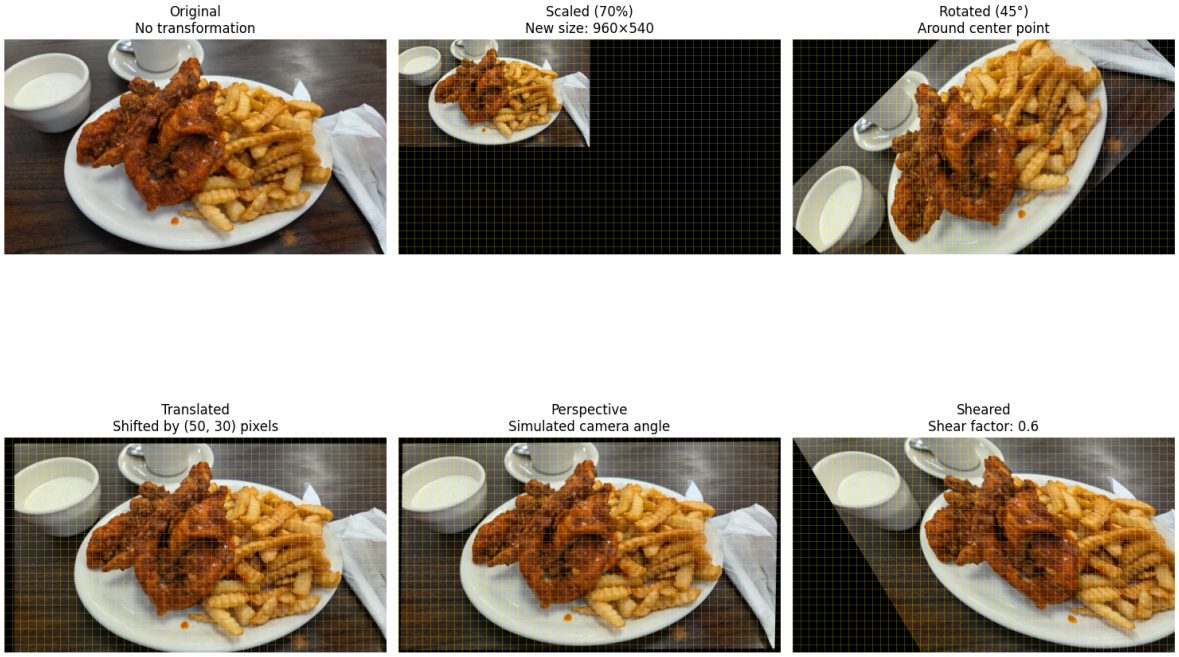



```
CHANNEL STATISTICS:
Red channel - Min:   1, Max: 255, Mean: 128.4
Green channel - Min:   0, Max: 255, Mean: 107.4
Blue channel - Min:   0, Max: 255, Mean: 86.9

Notice how different parts of the image contribute different amounts to each
```

transforming the image

```
# Use our original color image for geometric transformations
img_for_transform = img_rgb.copy()
height, width = img_for_transform.shape[:2]

# 1. Scaling (resize) Changed code for size to see a more drastic change
scale_factor = 0.5
new_width = int(width * scale_factor)
new_height = int(height * scale_factor)
scaled_img = cv2.resize(img_for_transform, (new_width, new_height), interpolat

# 2. Rotation Changed code to review angle change.
rotation_angle = 45 # degrees
center = (width // 2, height // 2)
rotation_matrix = cv2.getRotationMatrix2D(center, rotation_angle, 1.0)
rotated_img = cv2.warpAffine(img_for_transform, rotation_matrix, (width, heigh

# 3. Translation (shifting)changed code here to review changes
tx, ty = 50, 30 # shift by 50 pixels right, 30 pixels down
translation_matrix = np.float32([[1, 0, tx], [0, 1, ty]])
translated_img = cv2.warpAffine(img_for_transform, translation_matrix, (width,

# 4. Perspective transformation (simulating camera angle)
# Define source points (corners of original image)
src_points = np.float32([[0, 0], [width, 0], [width, height], [0, height]])
# Define destination points (perspective effect)
dst_points = np.float32([[20, 30], [width-10, 20], [width-30, height-20], [10,
perspective_matrix = cv2.getPerspectiveTransform(src_points, dst_points)
perspective_img = cv2.warpPerspective(img_for_transform, perspective_matrix, (

# 5. Shearing (skewing)
shear_factor = 0.6
shear_matrix = np.float32([[1, shear_factor, 0], [0, 1, 0]])
sheared_img = cv2.warpAffine(img_for_transform, shear_matrix, (width, height))

# Display all transformations
plt.figure(figsize=(15, 12))

transformations = [
```

```
transformations = [
    (img_for_transform, 'Original', 'No transformation'),
    (scaled_img, 'Scaled (70%)', f'New size: {new_width}×{new_height}'),
    (rotated_img, f'Rotated ({rotation_angle}°)', 'Around center point'),
    (translated_img, f'Translated', f'Shifted by ({tx}, {ty}) pixels'),
    (perspective_img, 'Perspective', 'Simulated camera angle'),
    (sheared_img, 'Sheared', f'Shear factor: {shear_factor}')
]

for i, (img, title, subtitle) in enumerate(transformations):
    plt.subplot(2, 3, i+1)
    if img.shape[:2] != img_for_transform.shape[:2]: # Handle different sizes
        # Pad smaller images for consistent display
        pad_h = max(0, height - img.shape[0])
        pad_w = max(0, width - img.shape[1])
        img_padded = np.pad(img, ((0, pad_h), (0, pad_w), (0, 0)), mode='const
        plt.imshow(img_padded)
    else:
        plt.imshow(img)

    plt.title(f'{title}\n{subtitle}')
    plt.axis('off')

    # Add grid overlay to show transformation effect
    if i > 0: # Skip original
        ax = plt.gca()
        # Add some reference lines
        for y in range(0, height, 40):
            ax.axhline(y=y, color='yellow', alpha=0.3, linewidth=0.5)
        for x in range(0, width, 40):
            ax.axvline(x=x, color='yellow', alpha=0.3, linewidth=0.5)

plt.tight_layout()
plt.show()

# Show transformation matrices
print("TRANSFORMATION MATRICES:")
print("\nRotation Matrix (30°):")
print(rotation_matrix)
print("\nTranslation Matrix:")
print(translation_matrix)
print("\nShear Matrix:")
print(shear_matrix)

print("\nGeometric Transformations Summary:")
print(" • Scaling: Changes image size (interpolation needed)")
print(" • Rotation: Rotates around a center point")
print(" • Translation: Shifts image position")
print(" • Perspective: Simulates 3D viewing angle")
print(" • Shearing: Skews the image shape")
print("\nThese are essential for image registration and augmentation!")
```




```
TRANSFORMATION MATRICES:

Rotation Matrix (30°):
[[ 7.07106781e-01  7.07106781e-01 -1.00660172e+02]
 [-7.07106781e-01  7.07106781e-01  8.36984848e+02]]

Translation Matrix:
[[ 1.  0. 50.]
 [ 0.  1. 30.]]

Shear Matrix:
[[1.  0.6 0. ]
 [0.  1.  0. ]]

Geometric Transformations Summary:
  • Scaling: Changes image size (interpolation needed)
  • Rotation: Rotates around a center point
  • Translation: Shifts image position
  • Perspective: Simulates 3D viewing angle
  • Shearing: Skews the image shape

These are essential for image registration and augmentation!
```

Use these questions to review your work and prepare for the Adventure Quest assignment

assignment.

1. **Understanding** - What surprised you most about how images are represented and processed? The biggest surprise for me the layers that they use to represent the image. Like with Color having 3 layers/channels that come together and form it. I also found the matrixes of it interesting with seeing how pixel intensity can change it as well.
2. **Connections** - How do the operations you implemented relate to the AI tools discussed in class?

Learning how machines see the process information helps down the road when making tools for it. The structure of how it gets from a-z lets us know what is important to implement every time.

3. **Applications** - What real-world uses can you think of for these techniques? Image processing already used some of fun one's would be filters. Making images brighter adding effects etc. But also it can enhance images for x-rays comparing scans. Self driving cars use images to view and make critical decisions.
4. **Creativity** - Which combination of operations produced the most interesting effect, and why? I thought the Sobel edges were interesting in particular to see how Machines can highlight a specific features in the image and how it can be put together.
5. **Further Learning** - Which aspect of image processing would you like to explore in your Adventure Quest?

I think the filters or how tools like photoshop are made to make new images, drawings in general. Building a tool that can automate based on prior choices customized to the user.

Discussion Points

- How does understanding the math change your view of AI image tools? It's helpful to see how it uses the numbers to "see" the image and just by adjusting some values can drastically change it. By knowing how it does the change and views it I can build tools with this in mind.
- What are the trade-offs between traditional and AI-powered approaches? With traditional you will get consistent results since you manually set up the parameters. Since you are the person who does evertything though it costs you time whereas an AI tool can be a set it and forget it

time whereas an AI tool can be a set it and forget it.

- How might you combine traditional techniques with modern AI tools in a project? The AI could be used as a quick base knowing what the end user typically like and from there still use traditional to adjust.
